

Kathryn D. Huff

CONTACT INFORMATION	<i>Professor</i> <i>Dept. of Nuclear Engineering and Engineering Physics</i> <i>University of Wisconsin–Madison</i>	UW e-mail: katy.huff@wisc.edu
RESEARCH INTERESTS	Advanced nuclear reactors and fuel cycles, multi-physics simulation, energy systems analysis, scientific computation, nuclear energy policy.	
PHD	University of Wisconsin–Madison, NUCLEAR ENGINEERING • An Integrated Used Fuel Disposition and Generic Repository Model for Fuel Cycle Analysis • Advisor: Professor Paul P.H. Wilson	Aug 2008 – Aug 2013
BA	University of Chicago, PHYSICS • Celestial Gain Calibrations of QUIET Telescope Polarimeters	Aug 2004 – Jun 2008
RESEARCH AND PROFESSIONAL EXPERIENCE	University of Wisconsin–Madison, Madison, WI <i>Professor, Nuclear Engineering and Engineering Physics</i> <i>Department Chair, Nuclear Engineering and Engineering Physics</i>	July 2026 – Present July 2026 – Present
	University of Illinois at Urbana-Champaign, Urbana, IL <i>Associate Professor, Nuclear Plasma and Radiological Engineering</i> <i>Affiliate Faculty, National Center for Supercomputing Applications</i> <i>Affiliate Faculty, Computational Science and Engineering</i> <i>Director, Advanced Reactors and Fuel Cycles group.</i>	Nov 2021 – Jun 2026 Aug 2016 – Jun 2026 Aug 2018 – Jun 2026
	Amazon Web Services, Seattle, WA <i>Amazon Scholar, Data Center Planning and Delivery, Global Energy Team</i> <i>Part-time private advisory consulting.</i>	Aug 2024 – Present
	Office of Nuclear Energy, Department of Energy, Washington, DC <i>Assistant Secretary, Nuclear Energy</i> <i>Senior Advisor to the Secretary, Nuclear Energy</i> <i>Acting Assistant Secretary, Nuclear Energy</i> <i>Principal Deputy Assistant Secretary, Nuclear Energy</i> <i>Presidentially appointed, Senate Confirmed Official leading the Office of Nuclear Energy</i> <i>On extended Unpaid Leave of Absence from the University of Illinois.</i>	May 2022 – May 2024 Jan 2022 – May 2022 May 2021 – Jan 2022 May 2021 – Jan 2022
	University of Illinois at Urbana-Champaign, Urbana, IL <i>Blue Waters Assistant Professor</i> <i>Principal Investigator, Advanced Reactors and Fuel Cycles group.</i>	Aug 2016 – Sep 2021
	University of California - Berkeley, NE Dept., Berkeley, CA <i>Postdoctoral Scholar, Nuclear Science and Security Consortium</i> <i>Data Science Fellow, Berkeley Institute for Data Science</i> <i>Developing computational tools and multiphysics models for advanced reactor safety analysis.</i>	Sep 2013 – Jul 2016 Aug 2014 – Jul 2016
	Argonne National Laboratory, Argonne, IL <i>Laboratory Graduate Research Appointee, Used Fuel Disposition Campaign</i> <i>Developed a used fuel disposition and generic repository computational model.</i>	Jun 2011 – Aug 2013
	University of Wisconsin - Madison, NEEP Dept., Madison, WI <i>Graduate Research Assistant, Computational Nuclear Engineering Research Group</i> <i>Developed and applied CYCLUS, a nuclear fuel cycle systems analysis tool.</i>	Jun 2008 – Aug 2013
	Idaho National Laboratory, Idaho Falls, ID <i>Graduate Research Assistant, Systems Analysis Campaign</i>	Jun – Aug 2010

Developed software functions and requirements for the Fuel Cycle Simulator concept.

Kavli Institute For Cosmological Physics, Chicago, IL **Jan 2005 – Jun 2008**
Research Assistant, Laboratory for Astrophysics and Space Research
Programmed & machined instrumentation. Planned protocol for QUIET polarimeter calibration.

Universidad de Chile, Physics Dept., Santiago, Chile **Jun – Sep 2006**
Research Assistant, Chicago-Chile Research Exchange Program
Constructed and operated a far-from-equilibrium granular materials experiment.

Los Alamos Neutron Science Center, Los Alamos, NM **Jun – Sep 2004**
Research Assistant, LANSCE-3 **May – Aug 2003**
Applied digital filtration algorithms and MCNPX models to experimental data.

HONORS AND
AWARDS

NPPE, Students Award for Excellence in Undergraduate Teaching	2017, 2021, 2025
Fin-Erth Women in Climate Award	2024
Presidential Citation, American Nuclear Society	2024
Distinguished Public Service Award, American Nuclear Society (ANS)	2024
40 Under 40, Nuclear News	2024
Meritorious Service Award, United States Nuclear Industry Council	2024
Secretary's Achievement Award, Uranium Strategy Tiger Team, U.S. Dept. of Energy (DOE)	2024
Secretary's Achievement Award, Pathways to Commercial Liftoff Team, U.S. DOE	2023
Warren K. Sinclair Medal, National Council on Radiation Protection & Measurements	2023
Presidential Nomination & Senate Confirmation, Assistant Secretary for Nuclear Energy	2022
Stanley H. Pierce Award, UIUC Engineering Council	2019
American Nuclear Society, Oestmann Professional Women's Achievement Award	2017
AE3, Collins Scholars Program Graduate	2017
UIUC, Teachers Ranked as Excellent	F 2016, S 2020
American Nuclear Society, Young Member Excellence Award	2016
National Energy Research Scientific Computing Allocation, Senior Investigator	2015–2016
Data Science Fellowship, Berkeley Institute for Data Science, UC Berkeley	2014–2016
Nuclear Science and Security Consortium Postdoctoral Fellowship, UC Berkeley	2013–2016
DOE Office of Science Laboratory Graduate Appointment, Argonne National Lab	2011–2013
Roy G Post Foundation Nuclear Waste Management Graduate Scholarship	2011
John Randall Memorial Scholarship, American Nuclear Society FCWMD	2009
J.A McDeavitt Scholarship, University of Chicago, Chicago, IL	2007–2008
University Scholar Award, University of Chicago, Chicago, IL	2004–2008
Los Alamos Distinguished Student Performance Award, Los Alamos National Lab	2004

GRANTS
AWARDED

Nuclear Science and Security Consortium 4	Period: 2026–2031
Source: DOE-NNSA Office of DNN R&D	Award Total: \$25,000,000
Role: Consortium Associate Director, UIUC PI, Thrust Area Lead	Huff Allocation: ~\$625,000
EINSTEIN: Enhanced Integrated Nuclear Systems for Transmutation and Efficient Isolation of Nuclides	Period: 2025–2028
Source: DOE-ARPA-E NEWTON Program	Award Total: \$1,025,405
Role: Collaborator	Huff Allocation: ~\$250,000
Nuclear Science and Security Consortium 3¹	Period: 2021–2026
Source: DOE-NNSA Office of DNN R&D	Award Total: \$25,000,000
Role: Consortium Co-PI, UIUC PI, Thrust Area Lead	Huff Allocation: \$625,000
Evaluation of micro-reactor requirements and performance in an existing well-characterized micro-grid¹	Period: 2020–2022
Source: DOE-NEUP	Award Total: \$800,000
Role: Co-PI	Huff Allocation: \$265,000

¹PI-ship transferred to other leadership in May 2021 corresponding with unpaid leave of absence.

Enabling Load Following Capability in the Transatomic Power MSR¹	Period: 2018–2021
Source: ARPA - E - MEITNER	Award Total: \$999,694
Role: Principal Investigator	Huff Allocation: \$205,000
US Research Software Sustainability Institute (URSSI)	Period: 2017–2018
Source: NSF - OAC - SI2 - S2I2 Conceptualization	Award Total: \$499,999
Role: Senior Personnel	Huff Allocation: N/A
Dynamic Transition Analysis with TIMES	Period: 2018–2019
Source: I ² CNER	Award Total: \$76,359
Role: Co-PI	Huff Allocation: \$76,359
Investigation of Agricultural Uses of Nuclear Waste Heat	Period: 2017–2018
Source: Exelon	Award Total: \$151,257
Role: Co-PI	Huff Allocation: \$11,678
Consortium for Verification Technology	Period: 2015–2020
Source: DOE-NNSA Office of DNN R&D	Award Total: \$25,000,000
Role: Consortium Co-PI, UIUC PI, CVT Investigator	Huff Allocation: \$347,000
Consortium for Nonproliferation Enabling Capabilities	Period: 2014–2019
Source: DOE-NNSA Office of DNN R&D	Award Total: \$25,000,000
Role: Consortium Co-PI, UIUC PI, Thrust Area Lead	Huff Allocation: \$648,000
Collaborative, Open-Source Curriculum Development	Period: 2017–2018
Source: UIUC Strategic Instructional Innovations Program	Award Total: \$19,347
Role: Principal Investigator	Huff Allocation: \$13,000
REU Site: INCLUSION at U. Illinois	Period: 2017–2020
Source: NSF - ACI	Award Total: \$380,036
Role: Senior Personnel	Huff Allocation: N/A
Demand-Driven Cycamore Archetypes	Period: 2016–2019
Source: DOE, NEUP R&D	Award Total: \$800,000
Role: Co-PI	Huff Allocation: \$395,066

BOOKS

- [1] A. M. Scopatz and **K. D. Huff**. *Effective computation in physics: Field guide to research with python*. O'Reilly Media, Sebastopol, CA, 1st edition, May 2015. URL: <http://shop.oreilly.com/product/0636920033424.do>

BOOK CHAPTERS

- [2] **K. Huff**. Foreword: Global Need for Nuclear Energy New Directions in. In *Nuclear Energy Innovation and Opportunities in Fission and Fusion for Global Decarbonization*, volume 1. 1st edition edition, Jan. 2026. URL: <https://shop.elsevier.com/books/new-directions-in-nuclear-energy/foss/978-0-443-33471-9>
- [3] S. Gesing, M. Pierce, S. Marru, M. Zentner, **K. D. Huff**, S. Bradley, S. B. Cleveland, S. R. Brandt, R. Ramnath, K. Kee, M. Dahan, B. M. V. Martínez, W. C. Sepulveda, and J. J. S. Mondragón. Science Gateways and AI/ML: How Can Gateway Concepts and Solutions Meet the Needs in Data Science? In *Critical Infrastructure - Modern Approach and New Developments*. IntechOpen, Mar. 2023. URL: <https://www.intechopen.com/chapters/86501>, doi:10.5772/intechopen.110144
- [4] **K. Huff**. Chapter One - Economics of Advanced Reactors and Fuel Cycles. In H. Bindra, editor, *Storage and Hybridization of Nuclear Energy*, volume 1, pages 1–20. Science & Technology Books Elsevier, Inc., Cambridge, MA, United States, 1 edition, Jan. 2019. URL: <http://www.sciencedirect.com/science/article/pii/B9780128139752000016>, doi:10.1016/B978-0-12-813975-2.00001-6
- [5] **K. Huff**. Case Study: Cyclus Project. In J. Kitizes, F. Imamoglu, and D. Turek, editors, *The Practice of Reproducible Research*, volume 1. University of California Press, University of California, Berkeley, 1 edition, 2017
- [6] **K. Huff**. Lessons Learned. In J. Kitizes, F. Imamoglu, and D. Turek, editors, *The Practice of Reproducible Research*, volume 1. University of California Press, University of California, Berkeley, 1 edition, 2017

- [7] S. G. Dotson, M. Munk, and **K. D. Huff**. Demonstrating the Osier framework for energy system and nuclear fuel cycle optimization. *Annals of Nuclear Energy*, 230:112151, June 2026. URL: <https://www.sciencedirect.com/science/article/pii/S0306454926000393>, doi:10.1016/j.anucene.2026.112151
- [8] S. M. Park, **K. D. Huff**, and M. Munk. Hybrid SN-Diffusion Neutronics Method for Control Rod Modeling in Molten Salt Reactors. *Nuclear Science and Engineering*, 0(0):1–26, May 2026. doi:10.1080/00295639.2026.2667091
- [9] A. M. Bachmann, Z. Richter, **K. D. Huff**, and M. Munk. Effects of Uranium Impurities in Down-blended HEU on HTGR Performance. *Nuclear Technology*, 0(0):1–19, Jan. 2026. doi:10.1080/00295450.2025.2560690
- [10] L. Seifert, S. Shahbazi, S. Richards, M. Munk, and **K. Huff**. Accuracy of the Scaled Flux Method for Spatially Resolved Nuclide Transmutation in Flowing-Fuel Molten Salt Reactors. *Nuclear Science and Engineering*, 0(0):1–13, Sept. 2025. doi:10.1080/00295639.2025.2568258
- [11] L. Seifert, B. Betzler, W. Wieselquist, M. Jessee, M. Munk, and **K. Huff**. Delayed Neutron Precursor Group Parameter and Spectra Generation from Fast Fission of ²³⁵U in SCALE. *Nuclear Science and Engineering*, 0(0):1–16, Aug. 2025. doi:10.1080/00295639.2025.2525754
- [12] W. E. Kennedy, R. A. Meserve, K. A. Higley, **K. D. Huff**, C. T. Hanson, M. Ford, D. Schultheisz, T. Smith, B. Kugelmass, A. Abou-Jaoude, J. R. Lovering, J. D. Semancik, G. V. Cullen, J. Cheatham, P. F. Peterson, E. Redmond II, S. M. Mirsky, M. Mahowald, M. G. Houts, D. Perkins, R. Vaghetto, J. Duhig, and C. J. D. VanHorne-Sealy. National Council on Radiation Protection (NCRP) 2024 annual meeting:advanced and small modular nuclear power reactors. *Journal of Radiological Protection: Official Journal of the Society for Radiological Protection*, 44(4), Oct. 2024. doi:10.1088/1361-6498/ad7ec4
- [13] N. Thiollière, X. Doligez, M. Halasz, G. Krivtchik, I. Merino, B. Mouginot, A. V. Skarbeli, A. Hernandez-Solis, F. Alvarez-Velarde, F. Courtin, H. Druenne, M. Ernoult, **K. Huff**, M. Szieberth, B. Vermeeren, and P. Wilson. Impact of fresh fuel loading management in fuel cycle simulators: A functionality isolation test. *Nuclear Engineering and Design*, 392:111748, June 2022. URL: <https://www.sciencedirect.com/science/article/pii/S0029549322001029>, doi:10.1016/j.nucengdes.2022.111748
- [14] M. Turkmen, G. J. Y. Chee, and **K. D. Huff**. Machine learning application to single channel design of molten salt reactor. *Annals of Nuclear Energy*, 161:108409, Oct. 2021. doi:<https://doi.org/10.1016/j.anucene.2021.108409>
- [15] A. Chaube, A. Chapman, A. Minami, J. Stubbins, and **K. D. Huff**. The role of current and emerging technologies in meeting Japan’s mid- to long-term carbon reduction goals. *Applied Energy*, 304:117669, Dec. 2021. URL: <https://www.sciencedirect.com/science/article/pii/S0306261921010308>, doi:10.1016/j.apenergy.2021.117669
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- [19] E. A. Miernicki, A. L. Heald, **K. D. Huff**, C. S. Brooks, and A. J. Margenot. Nuclear waste heat use in agriculture: History and opportunities in the United States. *Journal of Cleaner Production*, 267:121918, Sept. 2020. URL: <http://www.sciencedirect.com/science/article/pii/S095965262031965X>, doi:10.1016/j.jclepro.2020.121918

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- [23] O. Ashraf, A. Rykhlevskii, G. Tikhomirov, and **K. D. Huff**. Whole core analysis of the single-fluid double-zone thorium molten salt reactor (SD-TMSR). *Annals of Nuclear Energy*, 137:107–115, Mar. 2020. URL: <http://www.sciencedirect.com/science/article/pii/S0306454919306255>, doi: <https://doi.org/10.1016/j.anucene.2019.107115>
- [24] M. Kamuda, J. Zhao, and **K. Huff**. A comparison of machine learning methods for automated gamma-ray spectroscopy. *Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment*, 954:161385, Feb. 2020. URL: <http://www.sciencedirect.com/science/article/pii/S0168900218313779>, doi:10.1016/j.nima.2018.10.063
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High-Temperature Reactor Commercial Power Plant. *Nuclear Technology*, 195(3):222–238, Sept. 2016. URL: <http://www.ans.org/pubs/journals/nt/a.38935>, doi:10.13182/NT16-2

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PROCEEDINGS [37] M. Fang, **K. Huff**, T. Kozlowski, J. F. Stubbins, L. Vergari, and A. D. Fulvio. EINSTein: enhanced integrated nuclear systems for transmutation and efficient isolation of nuclides. In *Hard X-Ray, Gamma-Ray, and Neutron Detector Physics XXVII*, volume 13621, pages 128–133. SPIE, Sept. 2025. doi:10.1117/12.3067982
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- [234] **K. D. Huff**, J. W. Bae, K. A. Mummah, R. R. Flanagan, and A. M. Scopatz. Current Status of Predictive Transition Capability in Fuel Cycle Simulation. In *Proceedings of Global 2017*, Seoul, South Korea, Sept. 2017
- [235] **K. D. Huff.** Keynote. In *Scientific Python Conference*, Austin, TX, United States, July 2017. URL: <https://scipy2017.scipy.org/>
- [236] **K. D. Huff.** Keynote. In *PyData 2016, Chicago, IL*, Chicago, IL, USA, Aug. 2016. URL: <https://pyvideo.org/pydata-2016/keynote-katy-huff.html>
- [237] **K. D. Huff.** A Computational Future for Science Education. In *Computational, Mathematics, Science & Engineering Seminar, Michigan State University*, East Lansing, MI, USA, Dec. 2015. URL: <https://katyhuff.github.io/2015-11-06-ncsa/>
- [238] **K. D. Huff.** Seminar. In *NNSA Nuclear Nonproliferation*, Argonne National Laboratory, Lemont, IL, Sept. 2017
- [239] **K. D. Huff.** Seminar. In *RPNSD Seminar*, Oak Ridge National Laboratory, Oak Ridge, TN, USA, June 2017
- [240] **K. D. Huff.** Seminar. In *Mechanical and Aerospace Engineering Seminar*, UC Davis, Davis, CA, USA, Apr. 2017
- [241] **K. D. Huff.** Lightning Talk. In *AE3 Lightning Symposium*, University of Illinois, Urbana-Champaign, IL, USA, Mar. 2017
- [242] **K. D. Huff.** Undergraduate Seminar. In *Nuclear, Plasma, & Radiological Engineering Seminar*, University of Illinois, Urbana-Champaign, IL, USA, Feb. 2017
- [243] **K. D. Huff.** Symposium. In *Berkeley Institute for Data Science Symposium*, UC Berkeley, Berkeley, CA, USA, Jan. 2017
- [244] **K. D. Huff.** Seminar. In *Informatics Seminar*, University of Illinois, Urbana-Champaign, IL, USA, Oct. 2016
- [245] **K. D. Huff.** Seminar. In *RPNSD Seminar*, Oak Ridge National Laboratory, Oak Ridge, TN, USA, Mar. 2016
- [246] **K. D. Huff.** Seminar. In *Nuclear Engineering Seminar*, University of Tennessee, Knoxville, Knoxville, TN, USA, Mar. 2016
- [247] **K. D. Huff.** Seminar. In *Nuclear, Plasma & Radiological Engineering Seminar*, University of Illinois, Urbana-Champaign, IL, USA, Dec. 2015
- [248] **K. D. Huff.** Keynote. In *Python in High Performance Computing Workshop, SC15*, Austin, TX, USA, Nov. 2015
- [249] **K. D. Huff.** Colloquium. In *NCSA Colloquium*, University of Illinois, Urbana-Champaign, IL, USA, Nov. 2015
- [250] **K. D. Huff.** Colloquium. In *Nuclear Engineering Colloquium*, North Carolina State University, Raleigh, NC, USA, Oct. 2015
- [251] **K. D. Huff.** Colloquium. In *Nuclear Engineering Colloquium*, Texas A&M University, College Station, TX, USA, Sept. 2015

- [252] **K. D. Huff.** Invited Colloquium. In *Mechanical & Nuclear Engineering Colloquium, Rensselaer Polytechnic Institute*, Troy, NY, USA, Sept. 2015
- [253] **K. D. Huff.** What Can Academia Learn from Open Source? - Panel. In *What Can Academia Learn from Open Source?*, University of Washington, Seattle, WA, USA, Feb. 2015. URL: <https://ti.to/github-events/academia-town-hall->

ENGINEERING
TEACHING

University of Illinois at Urbana-Champaign

DEPT. OF NUCLEAR, PLASMA, AND RADIOLOGICAL ENGINEERING

NPRE 100, *Introduction to Nuclear, Plasma, and Radiological Engineering*

Fall 2024

Fall 2025

NPRE 247, *Modeling Nuclear Energy Systems*

Fall 2018

NPRE 412, *Nuclear Power Economics and Fuel Management*

Fall 2016

Fall 2017

Spring 2020

Spring 2021

Spring 2025

Spring 2026

NPRE 446, *Radiation Interactions with Matter I*

Fall 2019

NPRE 555, *Reactor Theory I*

Spring 2018

Fall 2020

Fall 2025

NPRE 560, *Reactor Kinetics and Dynamics*

Spring 2019

Spring 2021

Fall 2024

GUEST
LECTURES

University of California, Berkeley, DEPT. OF NUCLEAR ENGINEERING

Nov 10, 2020

NE 100, *Introduction to Nuclear Engineering*

Nuclear Fuel Cycle, Advanced Reactors

University of California, Berkeley, DEPT. OF NUCLEAR ENGINEERING

Apr 1,3,22, 2015

NE 155, *Introduction to Numerical Simulations in Radiation Transport* Point Reactor Kinetics, Monte Carlo Methods

University of California, Berkeley, DEPT. OF NUCLEAR ENGINEERING

Sep 11, 2014

NE 255, *Numerical Simulation in Radiation Transport*

Best Practices in Computational Nuclear Engineering

University of Wisconsin - Madison, DEPT. OF NUCLEAR ENGINEERING

Apr 1&3, 2013

NE 571, *Economic and Environmental Aspects of Nuclear Energy*

Nuclear Waste Repository Technology, Policy, and History

University of Wisconsin - Madison, DEPT. OF NUCLEAR ENGINEERING

Sep 9&11, 2009

NE 406, *Nuclear Reactor Analysis*

UNIX Shell, Basic Scripting, Environment Variables, Permissions, Regular Expressions, Makefiles

University of Wisconsin - Madison, DEPT. OF NUCLEAR ENGINEERING

Feb 10, 2010

NE 506, *Practicum in Monte Carlo Radiation Transport*

UNIX Shell, Basic Scripting, Environment Variables, Permissions, Regular Expressions, Makefiles

INVITED
SCIENTIFIC
COMPUTING
TEACHING

SciPy Conference, Austin, TX

Jul 6–7, 2015

Introductory Python For Scientific Software

University of Split , Split, Croatia G-Node Advanced Scientific Programming in Python Summer School	Sep 8–13, 2014
SciPy Conference , Austin, TX Version Control and Unit Testing For Scientific Software	Jun 25, 2013
University of Chicago, Graduate School , Chicago, IL Computational Literacy Workshop	Jan 12–13, 2013
University of California, Berkeley , Berkeley, CA Department of Statistics Scientific Computing Workshop	Oct 20–21, 2012
Lawrence Berkeley National Laboratory , Berkeley, CA Software Carpentry Python Workshop	Oct 17–18, 2012
International Center for Theoretical Physics , Trieste, Italy UNESCO/IAEA Advanced School on Scientific Software Development	Feb 20–Mar 2, 2012
University of Toronto , Toronto, ON, Canada SciNet Consortium For High Performance Computing Software Carpentry Bootcamp	Nov 7–8, 2011
American Nuclear Society Winter Meeting , Washington, D.C. Young Professionals Congress Hacker Within Scientific Computing Tutorial	Nov 1, 2011
Michigan State University , East Lansing, MI Institute for Cyber Enabled Research (iCER) and BEACON Center THW Bootcamp	Jun 4–5, 2011

SCIENTIFIC COMPUTING TEACHING

Berkeley Institute for Data Science , Berkeley, CA Managing Databases in SQL	Jan 14–15, 2015
Berkeley Institute for Data Science , Berkeley, CA Testing for Scientific Software	Jun 4–5, 2015
Lawrence Berkeley National Laboratory , Berkeley, CA Women in Science and Engineering Bootcamp	Apr 14–15, 2014
The University of Chicago , Chicago, IL Software Carpentry Scientific Computing Workshop	Apr 2–3, 2012
The University of Wisconsin , Madison, WI The Hacker Within Software Carpentry Bootcamp	Jan 12–14, 2011
The University of Wisconsin , Madison, WI The Hacker Within Python Bootcamp	Jan 12–14, 2010
The University of Wisconsin , Madison, WI The Hacker Within C++ Bootcamp	Mar 24–31, 2009
The University of Wisconsin , Madison, WI University of Wisconsin, Hacker Within UNIX Bootcamp	Jan 12–15, 2009

POSTDOCTORAL RESEARCHERS

<u>NAME</u>	<u>DATES</u>	<u>ROLE</u>
Mehmet Turkmen	2019–2020	Advisor
Alexander Lindsay	2016–2017	Advisor

GRADUATE RESEARCHERS

<u>NAME</u>	<u>DEGREE - YEAR</u>	<u>ROLE</u>
Michael Cheng	MS - 2017	MS Second Reader
Mark Kamuda	MS - 2017	MS Second Reader
Mark Kamuda	PhD - 2019	PhD Advisor
Gregory Westphal	MS - 2019	MS Advisor
Jin Whan Bae	MS - 2019	MS Advisor
Erik Medhurst	MS - 2020	MS Advisor
Andrei Rykhlevskii	PhD - 2020	PhD Advisor

Katherine C. Hepler	PhD - 2020	Dissertation Committee Chair
Alvin Lee	MS - 2020	MS Second Reader
Roberto Fairhurst-Agosta	MS 2020	PhD Advisor
Anshuman Chaube	MS - 2021	PhD Advisor
Gwendolyn Chee	PhD - 2022	PhD Co-Advisor
Amanda Bachmann	PhD - 2023	PhD Co-Advisor
Sun Myung Park	PhD - 2024	PhD Advisor
Nathan Ryan	MS - 2024	MS Advisor
Samuel Dotson	PhD - 205	PhD Advisor
Zoë Richter	PhD - (est. 2026)	PhD Advisor
Luke Seifert	PhD - (est. 2026)	PhD Advisor
Oleksandr Yardas	PhD - (est. 2026)	PhD Advisor
Elijah Capps	PhD - (est. 2029)	PhD Advisor
Liam Pohlmann	PhD - (est. 2029)	PhD Advisor
Samar Abdelkawy	MS - (est. 2026)	MS Advisor

UNDERGRADUATE RESEARCHERS	<u>NAME</u>	<u>DEGREE - YEAR</u>	<u>SCHOLARSHIPS</u>
	Jin Whan Bae	BS - 2017	NPPE Outstanding Undergrad Research ANS Best Student Fuel Cycle Presentation
	Kathryn Mummah	BS - 2017	Roy G. Post Foundation Scholarship ANS FCWMD Randall Scholar
	Eric Riewski	BS - 2017	
	GyuTae Park	BS - (est. 2018)	
	Yukun Tan	BS - (est. 2018)	Students Pushing Innovation
	Lu Kissinger	BS - 2019	
	Xin Wen	BS - 2018	Students Pushing Innovation
	Daniel Chiu	BS - 2020	
	Tyler Kennelly	BS - 2019	
	Bradley Ellis	BS - 2019	
	Adam Pichman	BS - 2019	
	Zoë Richter	BS - 2018	
	Gavin Davis	BS - 2021	
	Kip Kleimenhagen	BS - 2021	
	David Atwater	BS - 2021	
	Nathan Ryan	BS - 2022	
	Anna Balla	BS - 2021	
	Nataly Panczyk	BS - 2024	
	Bella Pequette	BS - 2025	
	Cesar Zambrano	BS - 2025	
	Rhys MacMillan	BS - (est. 2026)	
	Owen Strong	BS - (est. 2028)	
	Matthew Disimone	BS - (est. 2027)	
	Bryan Park	BS - (est. 2027)	
	Thomas Bohnert	BS - (est. 2028)	
	Madeleine Wallman	BS - (est. 2028)	

VISITING RESEARCHERS	<u>NAME</u>	<u>DATES</u>	<u>LEVEL - INSTITUTION</u>
	Gavin Ridey	2017	BS–University of Tennessee, Knoxville
	Aditya Bhosale	2017	BS - IIT, Bombay
	Snehal Chandan	2017	BS - IIT, Bombay
	Eleonora Skrzypek	2019	PhD - Warsaw University of Technology, Poland

SCIENTIFIC COMPUTING SKILLS	Languages	bash/csh, C++, FORTRAN, Perl, Python, XML
	Build Systems	make, CMake, automake
	Databases	HDF5, SQL

	Test Frameworks	CTest, GoogleTest, nose
	Version Control	cvs, git, hg, svn
	Other Tools	Doxygen, Sphinx, GoldSim, L ^A T _E X, Mathematica, MatLab, MCNP, MOOSE
EDITING AND REVIEWING	Editorial Board	<i>Nuclear Technology</i> 2024 – present
		<i>Journal of Open Source Education</i> 2018 – 2021
		<i>Journal of Open Source Software</i> 2016 – 2021
		<i>Nuclear Technology</i> 2018 – 2021
		<i>Nuclear Engineering and Design</i> 2020 – 2021
		<i>Papers in Physics</i> 2020 – 2023
		<i>Proceedings of the SciPy Scientific Python Conference</i> 2013, 2015, & 2017
		<i>Annals of Nuclear Energy</i>
		<i>Journal of Nuclear Energy Science and Power Generation Technology</i>
		<i>Nuclear Engineering and Design</i>
	Manuscript Referee	<i>Nuclear Science and Engineering</i>
		<i>Nuclear Technology</i>
		<i>Progress in Nuclear Energy</i>
	Grant Proposal Referee	<i>Dept. of Energy Nuclear Energy University Programs</i>
		<i>Dept. of Energy Technology Commercialization Fund</i>
		<i>Blue Waters Fellows Program</i>
		<i>Alfred P. Sloan Foundation</i>
	Book Proposal Referee	<i>O'Reilly Media</i>
		<i>Elsevier</i>
PROFESSIONAL SERVICE	Member , Argonne Board of Governors Program Review for Nuclear Energy	2025–present
	Member , Advocacy Council, Nuclear Matters, Washington D.C.	2025–present
	Member , External Affairs Committee, ANS	2025–present
	Vice Chair , Mathematics and Computation Division, ANS	2025–present
	Advisory Committee , Digital Information Technology, Sloan Foundation	2019–2021
	Chair , Nonproliferation and Policy Division, ANS	2020–2021
	Executive Committee , Mathematics and Computation Division, ANS	2020–2021
	Vice Chair , Nonproliferation and Policy Division, ANS	2019–2020
	Chair & Host , Technical Workshop on Fuel Cycle Simulation	2019
	Past Chair (<i>ex officio</i>), Fuel Cycle & Waste Management Division, ANS	2016–2017
	Co-Organizer , Technical Workshop on Fuel Cycle Simulation	2017
	Technical Program Committee , IHLRWM Conference	2017
	Chair , Fuel Cycle & Waste Management Division, ANS	2016–2017
	Vice Chair , Fuel Cycle & Waste Management Division, ANS	2015–2016
	Chair , Steering Committee, Software Carpentry Foundation	2014–2015
	Secretary–Treasurer , Fuel Cycle & Waste Management Division, ANS	2013–2015
	Secretary , Young Members Group, ANS	2013–2014
	Technical Program Co-Chair , SciPy, Scientific Python Conference	2013–2014
	Member , Next Generation Leadership Committee, Waste Management Symposium	2013–2014
	Moderator, Organizer, Panelist , inSCIght Scientific Computing Podcast	2011–2013
	Co-Founder , Nuclear Pride, LGBTQA Organization	2011–2013
DEPARTMENTAL SERVICE	Co-Founder, Treasurer, President , Hacker Within Scientific Computing Group	2008–2011
	Governor, Treasurer , University of Wisconsin ANS student section	2008–2010
	Graduate Committee , Qualifying Exam Sub-Committee	2024–present
	Graduate Committee , Admissions Sub-Committee	Spring 2024

	Faculty Advisor, UIUC ANS Student Section	2024–present
	Faculty Advisor, UIUC ANS Student Section	2016–2021
	Undergraduate Committee,	2019–2021t
	Graduate Committee, Qualifying Exam Sub-Committee	2017-2019
	Graduate Committee, Admissions Sub-Committee	Spring 2017
	Graduate Committee, Admissions Sub-Committee	Fall 2016
	Advisory Committee,	2017–2018
	Faculty Search Committee,	2017–2018
	Faculty Advisor, UIUC WiN Student Section	2017–2018
COLLEGE SERVICE	Member, Courses and Curriculum Committee	2025–present
	Fellow, Education Innovation Fellows	2025–present
	Member, Instructional Facility Working Group	2017–2018
	Selection Committee, Clare Boothe Luce (CBL) Research Scholars	2020–2021
	Member, Engineering IT Governance Education Working Group	2020–2021
	Faculty Mentor, ARISE program	2019–2020
	Member, ENG/TE Liaison Committee	2020–present
	Member, Instructional Facility Working Group	2017–2018
	Faculty Advisor, UIUC CSE The Hacker Within Scientific Computing Group	2016–2017
CAMPUS SERVICE	Senator, U. Illinois at Urbana-Champaign Faculty Senate	2025–present
	Steering Committee Member, Illinois Data Science Initiative	2018
	Hack Mentor, Hack Illinois	2017
CONSULTING	Amazon Web Services Seattle, WA	2024–Present
	<i>Amazon Scholar</i> Nuclear energy due dilligence, data center planning and delivery	
	Thomas Edison State University Trenton, NJ	2018–2019
	<i>Subject Matter Expert</i> Institute of Nuclear Power Operations (INPO) Academic Program Review	